CITY AND COUNTY OF DENVER
DEPARTMENT OF TRANSPORTATION AND INFRASTRUCTURE

PROJECT SPECIAL PROVISIONS

ALAMEDA LANE REDUCTION

2024-PROJMSTR-0000292

add construction duration o
days

GENERAL CONTRACT CONDITIONS

General Contract Conditions shall be the City and County of Denver, Department of Public Works "Standard Specifications for Construction General Contract Conditions," 2011 or revised current edition and hereinafter modified for this project.

STANDARD CONSTRUCTION SPECIFICATIONS

Standard Construction Specifications for this project shall be the Colorado Department of Transportation (CDOT) "Standard Specifications for Road and Bridge Construction" adopted in 2023 and as hereinafter modified. Sections 100 through 109 (Division 100) of the CDOT "Standard Specifications for Road and Bridge Construction" contain General Provisions specific to CDOT and may conflict with the City and County of Denver "Standard Specifications for Construction, General Contract Conditions," 2011 Edition.

Sections 103 and 108 of the CDOT "Standard Specifications for Road and Bridge Construction" are hereby deleted. In the event of any conflicting Division 100 General Provisions of the CDOT "Standard Specifications for Road and Bridge Construction", the CCD "Standard Specifications for Construction, General Contract Conditions," 2011 Edition shall prevail and apply to this contract.

The following Project Special Provisions supplement or amend the referenced "Standard Specifications for Road and Bridge Construction" adopted in 2023 by the Colorado Department of Transportation.

DETAILED CONSTRUCTION SPECIFICATIONS

The Detailed Specifications for this project are defined in the City of Denver Contract Requirements and Contract Documents.

STANDARD CONSTRUCTION DETAILS

It is the intent of the City to use applicable details from the Colorado Department of Transportation M&S Standards, the City of Denver's Wastewater Management Division Standard Details, and the Denver Water Department's Standard Drawings, most recent editions. Drainage-related appurtenances shall follow the City and County of Denver (CCD) Department of Public Works document titled "Storm Drainage Design and Technical Criteria". This document, along with the City's 2017 Transportation Standards and Details for the Engineering Division, can be found at the following web address:

https://www.denvergov.org/content/dam/denvergov/Portals/705/documents/guidelines/PWES-001.5-Transportation_Standards_and_Details_for_the_Engineering_Division.pdf

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PROJECT SPECIAL PROVISIONS
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The Colorado Department of Transportation's Standard Specifications for Road and Bridge Construction, dated 2023, controls construction of this project. The following Special Provisions supplement or modify the Standard Specifications and take precedence over the Standard Specifications and Plans.

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Need to add Revision of Section 614 for RECTANGULAR RAPID FLASHING BEACON and PEDESTRIAN PUSH BUTTON POST ASSEMBLY from the CTN-2 go by doc (see attached)

CDOT STANDARD SPECIAL PROVISIONS

Name	Date	No. of Pages
Revision of Section 106 – Conformity to the Contract of Hot Mix Asphalt (less than 5,000 tons)	(January 8, 2024)	4
Revision of Section 109 – Measurement and Payment	(December 9, 2022)	1
Revision of Sections 612 and 713 – Delineators and Reflectors	(December 9, 2022)	4
Affirmative Action Requirements – Equal Employment Opportunity	(October 1, 2022)	13

CONSTRUCTION LIMITS

The construction limits for this project are defined as along Alameda Avenue from the intersection of Lincoln Street (west) to Franklin Street (east). Furthermore, the construction limits shall include local streets as necessary to complete the work and to facilitate pedestrian and traffic routing during construction including placement of signage on any street to be used for construction detours within the Contract Drawings.

All work shall occur within the City and County of Denver right-of-way, City and County of Denver Park Space, CDOT right-of-way, or within easements acquired for the project. Impacts to existing landscaping, concrete, asphalt, sidewalk, structures, sprinklers, curbs, gutters, or any other elements that occur beyond the right-of-way shall be repaired by the Contractor at their own expense. Impacts to existing landscaping, fencing, concrete, asphalt, sidewalk, structures, sprinklers, curbs, gutters, or any other elements that occur within the right-of-way but are not specifically designated to be removed by the Contract Drawings or by the Engineer, shall be repaired by the Contractor at their own expense.

COMMENCEMENT AND COMPLETION OF WORK

(Working Hours and Schedule)

The Contractor shall commence work under the Contract per the City General Contract Conditions. The Contractor shall complete all work within 365 calendar days in accordance with the "Notice to Proceed." The minimum salient features shall be shown on the Contractor's Critical Path Schedule:

change to 180

1. Notice to Proceed
2. Mobilization
3. Public Information Services
4. Construction Traffic Control
5. Resets and Adjustments
6. Curb
7. HMA Paving
8. Signal Work
9. Signing and Striping
10. Project Cleanup
11. Anticipated Substantial Completion
12. Anticipated final acceptance

Contractor Schedule

The Contractor's progress schedule shall be a Critical Path Method Schedule. The schedule shall include 30 days of down time (float) for utility conflicts and delays. The Contractor will not be compensated for administrative or management costs incurred during down time for utility or other delays. No additional payment will be made for out of sequence work.

Work Restrictions for City Holidays and Mandatory City Furlough Days

Work shall not occur on holidays listed below. Work shall cease at 12:00 p.m. (noon) the day before the holiday or on the Friday before a Monday holiday. Work may resume at 8:00 a.m. the day after the holiday or the Monday after a Thursday and/or Friday holiday. All travel ways shall be open to the public and clear of obstructions during the holiday non-working periods. Holidays include:

- New Year's Day
- Martin Luther King Jr. Day
- Presidents' Day/Washington-Lincoln Day
- Cesar Chavez Day
- Memorial Day
- Juneteenth
- Independence Day
- Labor Day
- Veterans Day
- Thanksgiving Day
- Christmas Day

Work Restrictions for City sponsored or sanctioned events

Work may be restricted by City sponsored or sanctioned events. The Engineer will inform the Contractor of such events a minimum of two weeks prior to any event.

Work Restrictions for City Noise Ordinance and Street Occupancy Permit

Work shall be in conformance with the City's Noise Ordinance and the Contractor's street occupancy permit. The contractor will be required to submit a noise ordinance permit for the scheduled weekend closures and for any additional night work that may get approved by the Engineer.

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COMMENCEMENT AND COMPLETION OF WORK
(Working Hours and Schedule)

Substantial Completion

Substantial completion will be granted for facilities open to the Public upon approval by the Engineer. Refer to Title 19 of the City's Standard Specifications for Construction General Contract Conditions, 2011 Edition for additional information.

REVISION OF SECTION 101
DEFINITION AND TERMS

Section 101 of the Standard Specifications is hereby revised for this project as follows:

Delete Subsection 101.10 and replace with the following:

101.10 Resident Engineer. The Department's Project Manager, or their designated representative.

Delete Subsection 101.18 and replace with the following:

101.18 Contract Modification Order. Change Order in accordance with the City & County of Denver's General Contract Conditions, Section 1102.

Delete Subsection 101.28 and replace with the following:

101.28 Department. The City and County of Denver Department of Transportation and Infrastructure (DOTI). A department within the City and County of Denver, Colorado.

Delete Subsection 101.29 and replace with the following:

101.29 Engineer. The Department's Project Manager, who is responsible for engineering and administrative supervision of the project.

Delete Subsection 101.37 and replace with the following:

101.37 Inspector. The Department Project Manager's authorized representative assigned to make detailed inspections of contract performance.

Delete Subsection 101.48 and replace with the following:

101.48 Preconstruction Conference. A meeting of DOTI project personnel as requested by the Department's Project Manager, Contractor project personnel and other stakeholders held prior to the beginning of construction at which topics pertinent to the successful prosecution of the work are discussed.

Delete Subsection 101.51 and replace with the following:

101.51 Project Engineer. The Department's Project Manager, or their designated representative.

Delete Subsection 101.58 and replace with the following:

101.58 Region Transportation Director. The Department's City Engineer, or their designated representative.

Delete Subsection 101.76 and replace with the following:

101.76 State. The City and County of Denver (as applicable) acting through its authorized representative.

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REVISION OF SECTION 101
DEFINITION AND TERMS

Subsection 101.97 shall be added as follows:

101.97 Project Manager. References to the "Project Manager" within the Project Special Provisions and Plans will refer to the Department's Project Manager.

Subsection 101.98 shall be added as follows:

101.98 Day. Where the word "day" is used without the reference "calendar" or "working" preceding it, it shall be interpreted as calendar day.

REVISION OF SECTION 102
BIDDING REQUIREMENTS AND CONDITIONS

Section 102 of the Standard Specifications are hereby revised for this project as follows.

All subsections except for 102.03 and 102.05 are hereby deleted.

Replace Subsection 102.05 with the following:

102.05 Examination of Plans, Specifications, Special Provisions, and Site of Work. The bidder is expected to examine the site of the proposed work, the proposal, plans, specifications, supplemental specifications, special provisions, and contract forms, before submitting a proposal. The submission of a proposal will be considered conclusive evidence that the bidder has made this examination and is aware of the conditions to be encountered in performing the work according to the Contract.

Supplemental design data is provided in AutoCAD DWG file formats. The supplemental data transmitted consists of:

(a) Project Alignments. Civil3D AutoCAD DWG files of all Proposed Geometry
(b) Surfaces. Civil3D AutoCAD DWG files of the following:
  (1) Existing Terrain Model
  (2) Finished Grade Terrain Model
(c) DWG files.
  (1) Existing Base File
  (2) Proposed Base File
  (3) Control Line Geometry

The Department does not warrant the supplemental data and this information is not considered to be a part of the Contract. If bidders use or modify the supplemental data in preparing a proposal or planning and prosecuting the work, all associated data is used at their own risk, and bidders are responsible for all conclusions, deductions, and inferences drawn from the original or modified supplemental data

REVISION OF SECTION 104
SCOPE OF WORK

Section 104 of the Standard Specifications is hereby revised for this project as follows:

All subsections except for 104.05 and 104.07 are hereby deleted.

REVISION OF SECTION 105
CONTROL OF WORK

Section 105 of the Standard Specifications is hereby revised for this project as follows:

Subsection 105.02 *shall include the following*

All certifications and test reports for all applicable submittals shall conform to sections 106.12 and 106.13.

Table 105-1
CONTRACTOR SUBMITTALS

Section No.	Description	Type	Contractor P.E. Seal Required?	Specialty Unit Review	Review Time (working days)
612	Tubular Marker (SM)(Surface Mount)	Product Sheet	No	Yes	10
614	RRFB	Product Sheet	No	Yes	10
627	Harden Centerline	Product Sheet	No	Yes	10
627	3' Long Rubberized Parking Blocks	Product Sheet	No	Yes	10
627	Modular Speed Bump/End Cap	Product Sheet	No	Yes	10

Subsection 105.09 shall be replaced by the following:

105.09 Coordination of Plans, Specifications, Supplemental Specifications, and Special Provisions. These specifications, the supplemental specifications, the plans, special provisions, and all supplementary documents are essential parts of the Contract, and a requirement occurring in one is as binding as though occurring in all. They are intendedto be complementary and to describe and provide for a complete work.

In case of discrepancy, the order of precedence is as follows:

a) Special Provisions
   1. Project Special Provisions (revisions to CDOT Standard Specifications)
   2. Standard Special Provisions (updates to CDOT Standard Specifications)
b) Plans/Drawings (Calculated dimensions will govern over scaled dimensions)
   1. Detailed Plans (Project-Specific Plans and Details)
   2. Transportation Standards and Details for the Engineering Division City and County of Denver Public Works Department
   3. City and County of Denver Wastewater Standard Details
   4. City and County of Denver Traffic Signal, Sign, and Pavement MarkingStandards Department of Transportation and Infrastructure
   5. CDOT Standard Plans
c) Special Contract Conditions City and County of Denver Department ofTransportation and Infrastructure
d) Denver Department of Aviation Department of Public Works Standard Specifications for Construction General Contract Conditions
e) CDOT Standard Specifications for Road and Bridge Construction
f) Rules and Regulations of the Executive Director of the Denver Departmentof Transportation and Infrastructure

The Contractor shall not take advantage of any apparent error or omission in the Contract. If the Contractor discovers an error or omission, the Engineer shall immediately be notified. The Engineer will make corrections and interpretations as necessary to fulfill the intent of the Contract.

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REVISION OF SECTION 105
COOPERATION BETWEEN CONTRACTORS

Subsection 105.12 shall include the following:
Other construction contractors may be working in the vicinity of the project. The Contractor shall conduct the work so as not to interfere with or hinder the progress or completion of the work being performed by other agencies or contractors. All traffic control conflicts that arise between the needs of the various construction contractors and other agencies shall be brought to the attention of the Engineer. The Engineer will decide the method of resolution.

All traffic control and detour conflicts that arise between the needs of the various construction contractors and other agencies shall be brought to the attention of the Engineer. The Engineer will decide the method of resolution.

Revise paragraphs 4,5, and 6 of subsection 105.20 as follows:

If damage occurs to an existing structure through improper maintenance per 105.19, the Contractor shall submit a repair procedure to the Engineer to repair the defect(s).

The repair categories and requirements are defined as follows:

a) *"In-kind" repairs*. In-kind repairs are repairs where the As-Built or Advertised plans are utilized to replace or repair damaged components with identical dimensions and materials used plans and where no plan modifications are made. In-kind repair procedures shall be reviewed and accepted by the Engineer before any repair. The use of approved repair grouts or doweled reinforcing with epoxy adhesive is permitted in in-kind repairs. Doweled reinforcing shall meet or exceed the strength requirements of the original design.

a) "*Modified repairs*". Modified repairs are those which deviate in dimensions and/or materials from the As-Built or Advertised plans or where plans are not available. Modified repair procedure submittals shall include calculations, independent design calculations, shop drawings, and/or working drawings per 105.02, and any other applicable section of the specifications for the needed repair. The Contractor's Engineer shall electronically seal Modified repair submittals.

Damage to new structures or modified structures, shall be repaired per the contract documents.

The Engineer of Record shall be notified and review all corresponding submittals before any repairs.

REVISION OF SECTION 106
CONTROL OF MATERIAL

Section 106 of the Standard Specifications is hereby revised for this project as follows:

Subsection 106.05 shall include the following:

For this project, Contractor process control testing of hot mix asphalt is mandatory. The contractor must prepare s Process Control Plan compliant with section 106.05 – Section A and provide it to the city at the preconstruction meetings. All work to put the plan together (and any revisions requested by the City), shall not be measured, and paid for separately but shall be included in the work of Mobilization (see section 626).

The sixth sentence of Subsection 106.08 is replaced with the following:

Stored materials will be paid in accordance with Title 9, GC 906 of the City's Standard Specifications for Construction General Contract Conditions, 2011 Edition.

REVISION OF SECTION 107
LEGAL RELATIONS AND RESPONSIBILITY TO PUBLIC

Section 107 of the Standard Specifications is hereby revised for this project as follows:

Subsections 107.04, 107.07 – 107.08, 107.11, 107.13-107.18 are hereby deleted.

Subsection 107.06 Safety, Health, and Sanitation Provisions shall include the following:

Contractors shall comply with the CCD Noise Control Ordinance for all construction activities scheduled to occur between the hours of 9:00 p.m. and 7:00 a.m. Monday – Friday and 5:00 p.m. and 8:00 a.m. Saturday through Sunday if noise levels are expected to exceed 50 dB(A) in residential areas or 60 dB(A) in commercial areas. Nighttime construction activities exceeding noise levels during these restricted hours may not be initiated without first obtaining this permit variance from CCD.

Under any circumstances where nighttime construction activities will take place between the restrictive hours listed above where noise levels will exceed the noise impact thresholds, the Contractor is required to apply for a construction noise variance no less than one month before the startup date for construction. Note: the CCD board only meets the 2nd Thursday of every month.

Applications shall be submitted through the Denver Department of Environmental Health (DEH) at a cost of $25.00 for application processing.

The Contractor is responsible for fulfilling the requirements set forth by CCD and is responsible for obtaining the permit variance prior to the construction startup date. All exemption requests and permitting requirements will not be paid separately but will be included in the work.

If the Contractor is ordered to cease operations due to violations of a noise ordinance as a result of equipment back-up alarms, the Contractor shall take whatever actions are necessary to comply with the ordinance and continue work on the project. This includes the option of using an observer in lieu of using the equipment's back-up alarms as allowed by 29 CFR 1926.601(b) (4) (II) of the OSHA Safety and Health Standards. Should the Contractor fail to mitigate the noise ordinance violation, the Contractor shall be deemed to have waived any right to a claim as a result of work suspension or being required to perform the work at times not specified in the Contract.

If the Contractor uses an observer in lieu of back-up alarms, the Contractor shall follow all of the OSHA requirements regarding the use of observers.

All costs incidental to the foregoing requirements shall be included in the original contract prices for the project.

Add subsection 107.06 immediately following subsection 107.06 as follows:

107.06 Performance of Safety Critical Work. The following work elements are considered safety critical work for this project:

- Work in heavy traffic conditions

The Contractor shall submit, for record purposes only, an initial detailed construction plan that addresses safe construction of each of the safety critical elements. When the specifications already require a plan, it shall be included as a part of this plan. The detailed construction plan shall be submitted two weeks prior to the safety critical element conference described below. The construction plan shall be stamped "Approved for Construction" and signed by the Contractor. The construction plan will not be approved by the Engineer.

The Construction Plan shall include the following:

- Safety Critical Element for which the plan is being prepared and submitted.

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REVISION OF SECTION 107
LEGAL RELATIONS AND RESPONSIBILITY TO THE PUBLIC

- Contractor or subcontractor responsible for the plan preparation and the work.
- Schedule, procedures, equipment, and sequence of operations, that comply with the working hour limitations
- Temporary works required: falsework, bracing, shoring, etc.
- Additional actions that will be taken to ensure that the work will be performed safely.
- Names and qualifications of workers who will be in responsible charge of the work:
    - Years of experience performing similar work
    - Training taken in performing similar work
    - Certifications earned in performing similar work
- Names and qualifications of workers operating cranes or other lifting equipment
    - Years of experience performing similar work
    - Training taken in performing similar work
    - Certifications earned in performing similar work
- The construction plan shall address how the Contractor will handle contingencies such as:
    - Unplanned events (storms, traffic accidents, etc.)
    - Structural elements that don't fit or line up
    - Work that cannot be completed in time for the roadway to be reopened to traffic
    - Replacement of workers who don't perform the work safely
    - Equipment failure
    - Other potential difficulties inherent in the type of work being performed
- Name and qualifications of Contractor's person designated to determine and notify the Engineer in writing when it is safe to open a route to traffic after it has been closed for safety critical work.
- Erection plan or bridge removal plan when submitted as required elsewhere by the specifications. Plan requirements that overlap with above requirements may be submitted only once.

A safety critical element conference shall be held two weeks prior to beginning construction on each safety critical element. The Engineer, the Contractor, the safety critical element subcontractors, and the Contractor's Engineer shall attend the conference.

After the safety critical element conference, and prior to beginning work on the safety critical element, the Contractor shall submit a final construction plan to the Engineer for record purposes only. The Contractor's Engineer shall sign and seal temporary works, such as falsework, shoring etc., related to construction plans for the safety critical elements. The final construction plan shall be stamped "Approved for Construction" and signed by the Contractor.

The Contractor shall perform safety critical work only when the Engineer is on the project site. The Contractor's Engineer shall be on site to inspect and provide written approval of safety critical work for which he provided signed and sealed construction details. Unless otherwise directed or approved, the Contractor's Engineer need not be on site during the actual performance of safety critical work, but shall be present to conduct inspection for written approval of the safety critical work.

When ordered by the Engineer, the Contractor shall immediately stop safety critical work that is being performed in an unsafe manner or will result in an unsafe situation for the traveling public. Prior to stopping work, the Contractor shall make the situation safe for work stoppage. The Contractor shall submit an acceptable plan to correct the unsafe process before the Engineer will authorize resumption of the work.

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REVISION OF SECTION 107
LEGAL RELATIONS AND RESPONSIBILITY TO THE PUBLIC

When ordered by the Engineer, the Contractor shall remove workers from the project that are performing the safety critical work in a manner that creates an unsafe situation for the public in accordance with subsection 108.06.

Should an unplanned event occur or the safety critical operation deviate from the submitted plan, the Contractor shall immediately cease operations on the safety critical element, except for performing any work necessary to ensure worksite safety, and provide proper protection of the work and the traveling public. If the Contractor intends to modify the submitted plan, they shall submit a revised plan to the Engineer prior to resuming operations.

All costs associated with the preparation and implementation of each safety critical element construction plan will not be measured and paid for separately, but shall be included in the work.

Nothing in the section shall be construed to relieve the Contractor from ultimate liability for unsafe or negligent acts or to be a waiver of the Colorado Governmental Immunity Act on behalf of the Department.

Prior to beginning work on the project, the contractor's surveyor shall preform a survey to verify all survey control points, City of Denver range points, section corners, and benchmarks as shown on the survey control diagram. The contractor shall be responsible for replacing all land survey monuments disrupted by construction activities or by negligence on the part of the contractor. The contractor is required to follow Colorado State Law regarding survey monuments. The contractor shall retain a Colorado Licensed professional land surveyor to reset any affected survey monuments. For further information contact the city and county of Denver Survey Department.

Subsection 107.12 Protection and Restoration of Property and Landscape shall include the following:

The Contractor shall save all existing vegetation (including trees, shrubs, ground covers, grasses, wetlands, & riparian) in this area, except for that vegetation which must be removed to accommodate construction of the project, per the plans. Specific areas of vegetation to be protected shall be as directed by the Project Manager and shall be protected by using orange construction fencing, or as noted in Section 212. Fencing for trees shall be installed in accordance with Section 212 or approved by the

Project Manager. Equipment shall not be installed and stockpile material shall not be stored within 15 feet of existing trees to remain.

The Contractor shall perform all the work in such a manner that the least environmental damage will result. All questionable areas or items shall be brought to the attention of the Project Manager for approval prior to removal or any damaging activity.

The Contractor shall promptly report any vegetation damaged or scarred during construction to the Project Manager for assessment of damages. Damaged or destroyed fenced vegetation shall be replaced at the expense of the Contractor. Vegetation of replaceable size shall be replaced at the Contractor's expense. When trees, shrubs beyond replaceable size, or wetlands have been damaged or destroyed, the Contractor shall be liable for the appraised value based upon the official current publications. For trees and shrubs, use the International Society of Arboriculture, Guide for Plant Appraisals. The Contractor shall pay any fines or jail time should a wetland be damaged, at no cost to the project. The value of disturbed vegetation shall be calculated according to the following formula:

(Vegetation size) x (Species) x (Location) x (Condition) x (Arborist or Wetland Specialist) = Vegetation value

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REVISION OF SECTION 107
LEGAL RELATIONS AND RESPONSIBILITY TO THE PUBLIC

A consulting Arborist retained by the City will determine the value of the trees and shrubs. A consulting Wetland Specialist shall determine the value of the wetland or wetland species. This value will be deducted from any money due to the Contractor.

The determination as to whether a plant is of replacement size or beyond will be made by the City's Landscape Architect or Wetland Specialist. Contact Ben Rickenbacker at CCD Forestry Department.

If the fence is knocked down or destroyed by the Contractor, the Project Manager will suspend the work, wholly or in part, until the fence is repaired to the Project Manager's satisfaction at the Contractor's expense. Time lost due to such suspension will not be considered a basis for adjustment of time charges but will be charged as contract time.

Subsection 107.19 Furnishing Right of Way shall include the following:

The City will be responsible for the securing of all necessary rights of way in advance of construction. All Temporary Construction Easements shall be No Work Areas. The Contractor shall not enter the No Work Areas established at the preconstruction meeting and shown on the plans before the possession date listed or until otherwise directed in writing by the Engineer. These areas are anticipated to include, but are not limited to, Goldrick Elementary School and the Gas Station at the southeast corner of Zuni & Mississippi. The contractor shall install temporary construction fencing in these areas as needed at the direction of the city. This fencing will not be measured and paid for separately but shall be included in the cost of mobilization (see section 626).

The City anticipates no delay towards the completion of the project due to the restrictions imposed herein.

Subsection 107.23 Archaeological and Paleontological Discoveries is hereby revised as follows:

When the Contractor’s operations, including materials pits and quarries, encounter plant or animal fossils, remains of prehistoric or historic structures, prehistoric or historic artefacts (bottle dumps, charcoal from subsurface hearths, old pottery, potsherds, stone tools, arrowheads, etc.), the Contractor’s affected operations shall immediately cease. The Contractor shall immediately notify the Project Manager, or other appropriate agency for contractor source pits or quarries, of the discovery of these materials. When ordered to proceed, the Contractor shall conduct affected operations as directed. Additional work, except that in contractor source materials pits or quarries, will be paid for by City and County of Denver as provided in the General Contract conditions when contract unit prices exist, or as extra work as provided in the General Contract Conditions when no unit prices exist. Delays to the Contractor, not associated with work in Contractor sources, because of the materials encountered may be cause for extension of contract time in accordance with the General Contract Conditions. If fossils, prehistoric or historic structures, or prehistoric or historic artefacts are encountered in a contractor source materials pit or quarry, all costs and time delays shall be the responsibility of the contractor.

REVISION OF SECTION 109
MEASUREMENT AND PAYMENT

Section 109 of the Standard Specifications is hereby revised for this project as follows:

Subsection 109.03, 109.06, 109.07, 109.09, 109.10 shall be deleted:

REVISION OF SECTION 202
REMOVAL OF STRUCTURES AND OBSTRUCTIONS

Section 202 of the Standard Specifications is hereby revised for this project as follows:

All equipment, labor, hauling, sorting, removal, delivery, and documentation of recyclable materials as per Special Contract Condition, "Greenprint Denver Requirements" will not be paid separately; all costs incurred by the Contractor to meet the requirements of Special Contract Conditions shall be included in the removal of the related appurtenance.

Subsection 202.02 General shall be revised to include the following:

The Contractor will mark the limits of removals in the field. The Project Manager or designee will verify these limits prior to commencement of removal efforts. Removals in the vicinity of trees to remain shall conform to applicable provisions of the Tree Retention and Protection specification.

The Contractor will report and follow the City and County of Denver Regulated Asbestos Contaminated Soil Standard Operating Procedures as applicable during the course of the project.

The Contractor shall clean each valve box of all foreign debris such that the operating nut of the valve is fully accessible to operation. The Engineer will obtain the Owner's inspection and written approval before accepting the work.

Subsection 202.03 Salvable Material is hereby replaced as follows:

All salvable material shown on the plans shall be removed, without unnecessary damage, in sections or pieces that may be readily transported and delivered by the Contractor to the location noted on the plans, or as directed by the Engineer. The Contractor shall be held responsible for the safekeeping of all salvable materials during the period of the Contract until they are delivered to the City. The Contractor shall make good or replace at his own expense any such materials damaged, stolen, or otherwise lost prior to receipt by the City. All salvable materials, as designated on the plans, shall remain the property of the City.

Subsection 202.04 shall include the following:

Surface areas disturbed by removal of ground signs shall be restored to a condition equal to or exceeding that before the sign was removed. In soft surfaces the entire sign, including the foundation shall be removed and the remaining hole is to be filled with Aggregate Base Course (Class 6) or other suitable material. In hard surface areas, such as sidewalks, the post is to be removed so that it is recessed a minimum of 1 inch below the finished surface, and the hole shall be patched with concrete or high strength grout.

Subsection 202.05 shall include the following:

Painting existing markings black, gray, or any other color shall not be considered an acceptable means of removing existing markings.

Subsection 202.07 Pavements, Sidewalks, Curbs is hereby replaced as follows:

All concrete pavement, sidewalks, structures, curbs, gutters, asphalt pavement, etc. designated for removal shall be broken into pieces and recycled or hauled to DADS.

Old concrete to be left in place and which abuts new construction, edges of pavement, sidewalks, curbs, etc., shall be sawcut to a true line with a vertical face.

Old asphalt to be left in place and which abuts new construction, edges of asphalt pavement, patching, etc., shall be wheel-cut or sawcut to a neat vertical face with minimal jagged edges to the satisfaction of the Project Manager. The Project Manager, at their discretion, may require asphalt to be sawcut.

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REVISION OF SECTION 202
REMOVAL OF STRUCTURES AND OBSTRUCTIONS

Removal of curb and gutter shall include the complete removal to the nearest joint. No payment will be made for sprinkler relocation, utility relocation, etc., beyond the limits of construction as shown on the plans.

Subsection 202.12 is hereby deleted and replaced as follows:

Traffic control to accomplish pavement marking removal is considered included in the cost for project-wide Construction Traffic Control and will not be measured and paid separately.

Payment for Removal of Ground Sign shall be full compensation for all work necessary to complete the item including filling of the hole from the footing removal with Aggregate Base Course (Class 6) and restoration of the disturbed area or recessing the post and filling in the hole with concrete or high strength grout.

The cost for recycling, including delivery, fees, and all labor, shall not be paid for separately, but will be included in the cost of the removal items. The cost for delivery of reused items shall not be paid for separately but shall be included in the cost of the removal items.

Removal of Asphalt will be measured by area in square yards, regardless of thickness. Any sawcutting required will not be measured and paid separately but shall be included in the cost of the work.

Removal of Sign Panel (Special)(Mast Arm Mounted) shall include all work to remove and dispose of the existing mast arm sign. This includes, but is not limited to removal of brackets, capping of any holes on the mast arm, and traffic control. The contractor shall not perform an work on the mast arm over active lanes of traffic unless otherwise approved by the City.

The accepted quantities will be paid at the contract unit price for each of the pay items that appear in the bid schedule. The work includes sawcutting, scraping, removing, hauling and disposal of such items, excavation, and subsequent backfill (with moisture/density control) to proposed subgrade elevation. The work shall also include salvageable materials removed, their custody, preservation, storage, haul, and disposal as provided herein.

Subsection 202.12 shall include the following:

Payment will be made under:

Pay Item	**Pay Unit**
Removal of Pavement Marking	Square Foot
Removal of Ground Sign	Each
Removal of Sign Panel	Each
Removal of Traffic Signal Head	Each
Removal of Sign Panel (Special)(Mast Arm Mounted)	Each

REVISION OF SECTION 208
EROSION CONTROL

Section 208 of the Standard Construction Specifications is hereby removed in its entirety and replaced with the following:

PART I: DEFINITIONS

Definitions used for this Section shall consist of those listed in Title 1 of the City and County of Denver "Standard Specifications for Construction, General Contract Conditions", 2011 edition.

Definitions used for this Section hereby incorporate those identified within the City and County of Denver Construction Activities Stormwater Manual (CASM).

Additional Definitions applicable to this Section are listed heretofore:

Basis of Payment: The terms under which "Work" is paid, as a designated "Pay Item" in accordance with the quantity measured and the "Pay Unit."

Best Management Practices (BMPs): Schedules of activities, prohibitions of practices, installation of devices, maintenance procedures, and other management practices deployed to stabilize the construction site to prevent or reduce the pollution of State Waters (see definition below). Stormwater BMPs can be classified as "structural" (i.e., devices installed or constructed on a site) or "non-structural" (procedures, such as modified landscaping practices).

Colorado Department of Health and Environment (CDPHE): State of Colorado, Water Quality Control Division responsible for issuance of State Construction Stormwater Permit.

Construction Activities Stormwater Discharge Permit (CASDP): Permit issued by the City for compliance with City & County of Denver Revised Municipal Code and Department of Public Works Rules & Regulations concerning the discharge of pollutants in storm generated runoff from construction sites to Municipal Separate Storm Sewer System (MS4, see definition below) or State Waters, via the Municipal Separate Storm Sewer System (MS4).

Construction Activities Stormwater Manual (CASM): City and County of Denver Construction Activities Stormwater Manual (CASM), 2010 edition.

Colorado Department of Transportation (CDOT): State agency that has published standards for Erosion Control with accompanying Erosion Control Supervisor certification courses.

Erosion Control Supervisor (ECS): The Erosion Control Supervisor is assigned by the Contractor to perform duties as described in this Section. The ECS shall be properly trained in BMPs per requirements of Part V below and shall be under the direction of a Professional Engineer licensed in the State of Colorado when performing any modifications to the Project Stormwater Management Plan (SWMP), as required by CDPHE.

Final Stabilization: Point of construction when all ground surface disturbing activities at the site have been completed and uniform vegetative cover has reached 70% of pre-disturbance vegetative cover (as judged by comparison to nearest fallow vegetation), or equivalent permanent features have been employed. At this point, all temporary BMPs can be removed, all construction and equipment maintenance wastes have been disposed of properly; and all elements of the Stormwater Management Plan have been completed.

Major SWMP Modification: Changes to the original SWMP that removes or adds additional area to the Project or modifies the hydrology or drainage of the Project. A Major SWMP Modification requires the submission of revised Stormwater Management Plan (SWMP) elements to the Permit Authority for review and approval. Any adjustments to a SWMP must be performed either by or under the direction of a Professional Engineer licensed in the State of Colorado.

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REVISION OF SECTION 208
EROSION CONTROL

Minor SWMP Modification: Modification to the SWMP that does NOT increase the scope or change hydrology of the Project but: modifies/improves specific BMPs in use at site, indicates progression in phasing of the Project, or specifies relocation of previously approved BMPs within the Project. Any adjustments to a SWMP must be performed either by or under the direction of a Professional Engineer licensed in the State of Colorado.

Municipal Separate Storm Sewer System (MS4): A conveyance or system of conveyances (including roads with drainage systems, municipal streets, catch basins, curbs, gutters, ditches, man-made channels, or storm drains):

a) owned or operated by a State, city, town, county, district, association, or other public body (created by or pursuant to State law) having jurisdiction over disposal of stormwater or other wastes, including special districts under State law such as a sewer district, flood control district or drainage district, or similar entity, or a designated and approved management agency under Section 208 of the Federal Clean Water Act that discharges to State Waters;
b) designed or used for collecting or conveying stormwater;
c) which is not a combined sewer; and
d) which is not part of a Publicly Owned Treatment Works (POTW).

Permit Authority: The Department authorized by the City to review and process CASDP Applications for Capital and/ or governmental sponsored Projects. The responsible City department serving as the Permit Authority is the Engineering, Regulatory and Analytics Office. As a clarification, the Development Services Department of the City serves as the point of intake and permit processing center.

Permit Enforcement Authority: The Department authorized by the City to inspect and enforce CASDP Rules and Conditions for all construction Projects within the City's MS4 Boundary. The responsible City department serving as the Permit Enforcement Authority is the Wastewater Management Division of the Department of Transportation and Infrastructure.

State Construction Stormwater Permit: Colorado Revised Statues require that all construction sites/development Projects, which, by definition, disturb one or more acres in area, shall be covered by a State issued general permit for construction activities. Information on the application requirements for the State permit can be obtained by phone at 303-692-3500; or by visiting their offices located at 4300 Cherry Creek Drive South, Denver, CO 80246 – 1530. or on the Web at:
https://www.colorado.gov/pacific/cdphe/news/water-quality-permits

State Waters: Any and all surface waters which are contained in or flow in or through this State, not to include waters in sewage systems, waters in treatment works of disposal systems, waters in potable water distribution systems, and all water withdrawn for use until use and treatment have been completed.

Examples of State Waters include, but are not limited to, perennial streams, intermittent or ephemeral gulches and arroyos, ponds, lakes, reservoirs, irrigation canals or ditches, wetlands, stormwater conveyances (when they discharge to a surface water), and groundwater.

Stormwater Management Plan (SWMP): The Stormwater Management Plan contains the requirements necessary to accomplish all the following:

The SWMP establishes a minimum standard to construct, install, maintain, and remove required BMPs during the life of the Contract to prevent or minimize pollution of stormwater due to erosion, sediment transport, and construction related pollutant generated during all phases of the Project. A SWMP consists of the following elements:

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REVISION OF SECTION 208
EROSION CONTROL

1. CASDP Narrative Worksheet with Narrative Report. The Narrative Report and supporting documents should fully address the methods to be used to prevent sediment, debris, and other pollutants from entering the MS4 and/ or State Waters in and around the Project area. Proposed structural and non-structural BMPs should be described with sufficient implementation detail to ensure that the logical phases of the proposed construction Project meet the performance standards listed in the CASM.
2. Proposed site drawings and Best Management Practice (BMP) installation details as they apply to the site conforming to the Urban Storm Drainage Criteria Manual, Vol. 3, "Best Management Practices", most current version as issued by the Mile High Flood District (MHFD), or those established by the City's Department of Transportation and Infrastructure. If erosion control drawings were included within the bid documents for the Project, they shall be used for bid purposes and initial planning/ deployment of BMPs on the Project. If provided drawings are signed/ sealed by a Professional Engineer, they have been pre-approved by the Permit Authority and may be used without revision for purposes of submitting for CASDP. If provided drawings do not have signature/ seal of Professional Engineer licensed by the State of Colorado, they will require revision by the Contractor with Professional Engineer signature/ seal prior to submission to the City and County of Denver for CASDP.
3. Supporting documentation related to proposed BMPs that are not currently identified in MHFD Vol. 3 or as otherwise published by the City.

Any preparation of or adjustments to a SWMP must be performed either by or under the supervision of a Professional Engineer licensed in the State of Colorado. SWMP elements submitted to the City shall also meet currently established criteria of the CDPHE as the SWMP must meet all local, State and Federal requirements.

Substantial Completion of Erosion Control: Point of construction when permanent BMPs have been installed, initial growth is in place, and the site is waiting for vegetative cover to reach 70% of pre-disturbance vegetative cover.

PART II: DESCRIPTION

This Work shall consist of constructing, installing, maintaining, and removing when required, BMPs during the life of the Contract until Final Stabilization to prevent or minimize erosion, sedimentation, and pollution of any waters including storm, drainageways, MS4, State Waters, and/ or wetlands. Work under this Section includes the Contractor obtaining required Permits, utilizing SWMP elements provided in the Contract, and/ or SWMP elements specifically prepared by the Contractor as defined herein. The work shall also consist of providing on-going maintenance and monitoring of the SWMP as may be necessary due to the specific and/or dynamic needs of the Project as well as meet all requirements set forth within the CASMP.

The Contractor shall coordinate the construction of temporary BMPs with the construction of permanent BMPs to assure economical, effective, and continuous erosion and sediment control and water pollution prevention throughout the construction period until Final Stabilization is achieved.

When a provision of this Section or an order by the Permit Enforcement Authority requires that an action be immediate or taken immediately, it shall be understood that the Contractor shall at once begin effecting completion of the action and pursue it to completion in a manner acceptable to the Permit Enforcement Authority, and in accordance with applicable Permitting requirements.

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REVISION OF SECTION 208
EROSION CONTROL

PART III: MATERIALS

The materials to be used for BMPs shall conform to each specific detail as set forth within the Project SWMP, as noted on the Contract Drawings, as required by the City or the State during any inspections or otherwise, and as proposed and used by the contractor in their best management practices.

PART IV: EROSION CONTROL PERMIT STATUS

The current SWMP status for the Project is as follows:

SCENARIO 1: CASDP is not required.

A SWMP is currently not required for this project as the proposed disturbed area and/ or proximity to stream does not meet the minimum criteria for requiring a CASDP. All provisions of the following Parts V-VIII of this Section are hereby deleted and shall not be made part of the Project. However, the responsibilities for minimizing sediment pollution from the Project have not been waived, and as such, the City hereby requires the Contractor to perform as specified in the following notes.

Because a SWMP and CASDP are not necessary for the Project, all cost for performance of the following notes shall not be paid for separately but are made a LS item (208-00200 Erosion Control Management) for the contractor to follow anticipated BMPs as described below. Note that this lump sum item will include however many concrete washouts the contractor deems necessary.

A CASDP Permit will not be required for this project, however, the Contractor and/or their authorized agents shall ensure that all potential pollutants generated during demolition, excavation, trenching, boring, grading, or other construction Work associated with this permit, be prevented from discharge to stormwater conveyance systems in the vicinity of the Project.

<u>The Contractor and/or their authorized agents</u> shall remove all sediment, mud, construction debris, or other potential pollutants that may have been discharged to or, accumulate in the flow lines of storm drainage appurtenances and public rights of ways of the City and County of Denver as a result of construction activities associated with this Project. All removals shall be conducted in a timely manner.

The Contractor shall be held responsible for remediation of any adverse impacts to the MS4, State Waters, waterways, wetlands, and or other public or private properties, resulting from work done as part of this project.

<u>The Contractor and/or their authorized agents</u> shall ensure that all loads of cut and fill material imported to or exported from the Project shall be properly covered to prevent loss of the material during transport on public rights of way." (Sec.49-552; Revised Municipal Code)

Approved erosion and sediment control 'Best Management Practices' shall be maintained and kept in good repair for the duration of the Project. All necessary maintenance and repair shall be completed immediately upon discovery of any deficiency or defect.

The Contractor and/or their authorized agents shall implement the following Best Management Practices (BMPs) on site during construction:

1. VEHICLE TRACKING CONTROL: This BMP is required at all access points to a construction site that are used by vehicular traffic or construction equipment.

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REVISION OF SECTION 208
EROSION CONTROL

2. INLET PROTECTION: This BMP is required on all existing or proposed storm sewer inlets in the vicinity of the construction site that may receive site runoff. The BMP must be appropriate to the type of storm inlet and appropriate for the ground surface at the inlet.

3. INTERIM SITE STABILIZATION: This BMP is required to provide a measure for preventing the discharge of sediment from construction sites where overlot grading or other site disturbance has occurred. This BMP is particularly necessary on sites where construction activities/disturbance will be limited to small areas of the project site. Acceptable BMPs include:

   a) Preserving existing vegetation
   b) Seeding and planting
   c) Mulching
   d) Mulching and seeding
   e) Temporary/Permanent re-vegetation operations
   f) Chemical soil stabilizer application (requires Permit Enforcement Authority approval)

4. WASTE MANAGEMENT/CONTAINMENT: This BMP requires that all construction wastes, fuels, lubricants, chemical wastes, trash, sanitary wastes, contaminated soils or debris shall be contained on site, protected from contact with precipitation or surface runoff, periodically removed from the construction site, and properly disposed of.

5. SPILL PREVENTION /CONTAINMENT: This BMP defines the measures proposed for preventing, controlling, or containing spills of fuel, lubricants, or other pollutants; and protecting potential pollutants from contact with precipitation or runoff.

6. CHUTE WASHOUT CONTAINMENT: Water used in the cleaning of ready mixed concrete truck delivery chutes shall be discharged into a predefined, bermed containment area on the job site. The required containment area is to be bermed so that wash water is totally contained. Wash water discharged into the containment area shall be allowed to infiltrate or evaporate. Dried concrete waste shall be removed from the containment area and properly disposed of.

   a) Should a predefined bermed containment area not be available due to the project size, or lack of an area with a suitable ground surface for establishing a containment area, proper disposal of ready mix washout and rinse off water at the job site shall conform to the approved techniques and practices identified in the Colorado Department of Public Health & Environment's training video entitled "Building For a Cleaner Environment, Ready Mix Washout Training", and its accompanying manual entitled, "Ready Mix Washout Guidebook, Vehicle and Equipment Washout at Construction Sites."
   b) The direct or indirect discharge of water containing waste concrete to the storm sewer system is prohibited (Sec.56-102a, c; Revised Municipal Code, City and County of Denver).
   c) Information about, or copies of the video and training manual are available from the Water Quality Control Division, Colorado Department of Public Health & Environment, 4300 Cherry Creek Drive South, Denver, Colorado 80222-1530, (303) 692-3555.

7. STREET SWEEPING: This BMP requires that paved surfaces which are adjacent to construction sites be swept in a timely manner when sediment and other materials are tracked or discharged on to them. Either sweeping by hand or use of street sweepers is acceptable. Street sweepers using water while sweeping is preferred in order to minimize dust. Flushing off paved surfaces with water is prohibited.

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REVISION OF SECTION 208
EROSION CONTROL

8. PERIMETER CONTROL: This BMP requires that a construction site install a perimeter control measure along the edge of the construction site, to prevent, or filter the discharge of surface runoff from the construction site. The type of perimeter control used shall be determined based on site conditions and location. Maintenance and repair of the control measure shall occur as needed, in a timely manner.
9. STOCK PILES: Soils that will be stockpiled for more than thirty (30) days shall be protected from wind and water erosion within fourteen (14) days of stockpile construction. Stabilization of stockpiles located within 100 feet of an MS4 or State Waters, or with slopes 3 to 1 or greater shall be completed within seven (7) days following stockpile construction. Stabilization and protection of the stockpile may be accomplished by any of the following: Mulching, Temporary/Permanent Revegetation Operations, Chemical Soil Stabilizer Application (requires Permit Enforcement Authority approval), or erosion control matting/Geotextiles. If stockpiles are located within 100 feet of an MS4 or State Waters, a drainageway or the site perimeter, additional sediment controls shall be required.
10. SAW CUTTING OPERATIONS: The Contractor shall protect all storm sewer facilities adjacent to any location where pavement cutting operations involving wheel cutting, saw cutting, or abrasive water jet cutting are to occur. The Contractor shall remove and properly dispose of all waste products generated by said cutting operations on a daily basis or as needed throughout the work day. The discharge of any water contaminated by waste products from cutting operations to the storm sewer system is prohibited. (Sec.56-102a, c; Revised Municipal Code, City and County of Denver).

PART VII: METHOD OF MEASUREMENT

Erosion Control Management will not be measured but will be paid for on a lump sum basis.

Removal of sediment & trash that is or is not generated by construction activities will not be measured separately but shall be included in the work. Any excavation required for the removal of sediment from traps, basins, areas adjacent to silt fences and erosion bales, and any other cleanout excavation of accumulated sediment, and removal of check dams or storm drain inlet protection will not be measured separately but shall be included in the work.

BASIS OF PAYMENT

Partial payments will be made:

90 percent upon installation
10 percent upon continued upkeep and removal

Payment will be made under:

Pay Item	**Pay Unit**
Erosion Control Management	Lump Sum

REVISION OF SECTION 210
RESET STRUCTURES

Section 210 of the Standard Specifications is hereby revised as follows:

Subsection 210.07 shall include the following:

Reset Sign Panel shall include the removal of the sign and all attachments. All attachments shall be inspected for wear prior to reuse. Worn attachments shall be replaced.

Subsection 210.13 shall include the following:

Payment will be made under:

Pay Item	**Pay Unit**
Reset Sign Panel	Each

All new attachments and connectors required to reset the sign in the new location shall be included in the cost of Reset Sign Panel.

REVISION OF SECTION 403
HOT MIX ASPHALT

Sections 403 of the Standard Specifications is hereby revised for this project as follows:

Hot mix asphalt (patching) shall be paid by the ton by batch ticket. Tack Coat will not be measured and paid for separately but shall be included in the cost of the work.

It is the Contractor's responsibility to provide Quality Control density testing and provide test results in a testing frequency, method and report as per Pre-construction submittal approved by the City Project Manager. Quality Control testing, reports and submittals will not be paid for separately but shall be included in the work.

Contractor shall confirm in writing test types, and frequency with CDOT and CCD prior to beginning work. Any materials/coordination required to support the City's IAT test shall not be measured and paid for separately but shall be included in the work. Additional details about the IAT testing will be provided at the preconstruction meeting.

Subsection 403.02 shall include the following:

The design mix for hot mix asphalt shall conform to the following:

TABLE 403-1

Property	**Test Method**	**Value For Grading**			
			S (100)	SX (100)	Patching
Air Voids, percent at: N (design)	CPL 5115		3.5 – 4.5	3.5 – 4.5	3.5 – 4.5
Lab Compaction (Revolutions): N (design)	CPL 5115		100	100	100
Stability, minimum	CPL 5106		30	30	30

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REVISION OF SECTION 403
HOT MIX ASPHALT

Aggregate Retained on the 4.75 mm (No. 4) Sieve for S, SX and SG, and on the 2.36mm (No. 8) Sieve for ST and SF with at least 2 Mechanically Induced fractured faces, % minimum*	CP 45		60	60	60
Accelerated Moisture Susceptibility Tensile Strength Ratio (Lottman), minimum	CPL 5109 Method B		80	80	80
Minimum Dry Split Tensile Strength, kPa (psi)	CPL 5109 Method B		205 (30)	205 (30)	205 (30)
Grade of Asphalt Cement, Top Layer			-	PG 76-28	PG 76-28
Grade of Asphalt Cement, Layers below Top			PG 64-22		PG 64-22
Voids in the Mineral Aggregate (VMA) % minimum	CP 48		See Table 403-2	See Table 403-2	See Table 403-2
Voids Filled with Asphalt (VFA), %	AI MS-2		65-75	65-75	65-75
Dust to Asphalt Ratio Fine Gradation Coarse Gradation	CP 50		0.6 – 1.2 0.8 – 1.6	0.6 – 1.2 0.8 – 1.6	0.6 – 1.2 0.8 – 1.6

Note: AI MS2 = Asphalt Institute Manual Series 2

Note: Mixes with gradations having less than 40% passing the 4.75 mm (No. 4) sieve shall be approached with caution because of constructability problems.

Note: Gradations for mixes with a nominal maximum aggregate size of one-inch or larger are considered a coarse gradation if they pass below the maximum density line at the #4 screen.

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REVISION OF SECTION 403
HOT MIX ASPHALT

Gradations for mixes with a nominal maximum aggregate size of 3/4" to 3/8" are considered a coarse gradation if they pass below the maximum density line at the #8 screen.

Gradations for mixes with a nominal maximum aggregate size of #4 or smaller are considered a coarse gradation if they pass below the maximum density line at the #16 screen.

*Fractured face requirements for SF may be waived by RME depending on project conditions.

All mix designs shall be run with a gyratory compaction angle of 1.25 degrees and properties must satisfy

Table 403-1. Form 43 will establish construction targets for Asphalt Cement and all mix properties at Air Voids up to 1.0 percent below the mix design optimum. CDOT will establish the production asphalt cement and volumetric targets based on the Contractor's mix design and the relationships shown between the hot mix asphalt mixture volumetric properties and asphalt cement contents on the Form 429. CDOT may select a different AC content other than the one shown at optimum on the Contractor's mix design in order to establish the production targets as contained on the Form 43. Historically, Air Voids adjustments typically result in asphalt cement increases from 0.1 to 0.5 percent. Contractors bidding the project should anticipate this change and factor it into their unit price bid.

Table 403-2

	Minimum Voids in the Mineral Aggregate (VMA)			
Nominal Maximum Size*, mm (inches)	*****Design Air Voids ****			
	3.5%	**4.0%**	**4.5%**	**5.0%**
37.5 (1½)	11.6	11.7	11.8	N/A
25.0 (1)	12.6	12.7	12.8	
19.0 (¾)	13.6	13.7	13.8	
12.5 (½)	14.6	14.7	14.8	
9.5 (⅜)	15.6	15.7	15.8	
4.75 (No. 4)	16.6	16.7	16.8	16.9
	* The Nominal Maximum Size is defined as one sieve larger than the first sieve to retain more than 10%. ** Interpolate specified VMA values for design air voids between those listed. *** Extrapolate specified VMA values for production air voids beyond those listed.			

The Contractor shall prepare a quality control plan outlining the steps taken to minimize segregation of HMA. This plan shall be submitted to the Engineer and approved prior to beginning the paving operations. When the Engineer determines that segregation is unacceptable, the paving shall stop and the cause of segregation shall be corrected before paving operations will be allowed to resume.

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REVISION OF SECTION 403
HOT MIX ASPHALT

Hot mix asphalt for the top lift of patching used on the roadway surface shall conform to the gradation requirements for Hot Mix Asphalt (Grading SX) (100) (PG 76-28). Hot mix asphalt for lower lifts of patching used on the roadway surface shall conform to the gradation requirements for Hot Mix Asphalt (Grading S) (100) (PG 64-22).

CDOT approved Warm Mix Asphalt (WMA) may be allowed on this project in accordance with CP 59. Unique requirements for WMA design, production and acceptance testing as documented during CDOT WMA approval shall be submitted and approved prior to creation of the Form 43 and before any WMA production on the project. Delays to the project due to WMA submittal and review will be considered within the Contractor's control and will be non-excusable.
Contractors proposing to use WMA shall supply detailed design, production and acceptance testing requirements prior to completion of the Form 43. Approved WMA submittals shall contain all of this information prior to CDOT approval. Only CDOT Approved WMA incorporating additives from the Approved Products List (APL) will be allowed for use on the project.

A minimum of 1 percent hydrated lime by weight of the combined aggregate shall be added to the aggregate for all hot mix asphalt.

Subsection 403.03 shall include the following:

If liquid antistripping additive is added at the plant, an approved inline blender must be used. The blender shall be in the line from the storage tank to the drier drum or pugmill. The blender shall apply sufficient mixing action to thoroughly mix the asphalt cement and antistripping additive.

The Contractor shall construct the work such that all roadway pavement placed prior to the time paving operations end for the year, shall be completed to the full thickness required by the plans. The Contractor's Progress Schedule shall show the methods to be used to comply with this requirement.

Delete subsection 403.05 and replace with the following:

403.05 The accepted quantities of hot mix asphalt will be paid for in accordance with subsection 401.22, at the contract unit price per ton for the bituminous mixture.

Payment will be made under:

Pay Item	**Pay Unit**
Hot Mix Asphalt (Patching) (Asphalt)	Ton

Aggregate, asphalt recycling agent, asphalt cement, additives, hydrated lime, and all other work and materials necessary to complete each hot mix asphalt item will not be paid for separately but shall be included in the unit price bid. When the pay item includes the PG binder grade, any change to the submitted mix design optimum asphalt cement content to establish production targets on the Form 43 will not be measured and paid for separately but shall be included in the work. No additional compensation will be considered or paid for any additional asphalt cement, plant modifications and additional personnel required to produce the HMA as a result in a change to the mix design asphalt cement content.

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REVISION OF SECTION 403
HOT MIX ASPHALT

Historically, typical asphalt cement increases reflected on the Form 43 are from 0.1 to 0.5 percent. However, the Contractor should anticipate the AC increases typical of his mixes. Contractors bidding the project should anticipate this change and factor it into their unit price bid.

When the pay item does not include the PG binder grade, asphalt cement will be measured and paid for in accordance with Section 411. Asphalt cement used in Hot Mix Asphalt (Patching) will not be measured and paid for separately but shall be included in the work.

Excavation, preparation, and tack coat of areas to be patched will not be measured and paid for separately but shall be included in the work.

REVISION OF SECTION 612
DELINEATORS AND REFLECTORS

Section 612 of the Standard Specifications is hereby revised for this project as follows:

Subsection 612.01 shall be revised to include the following:

This work consists of the installation and furnishing of tubular markers in accordance with these specifications and in conformity with the lines, grades and details shown on the plans or established.

Subsection 612.02 shall be revised to include the following:

(d) *Tubular Marker (City Post SM).* The City Post Surface Mount is a high-performance channelizer post system comprised of a tubular marker, a top cap and an integrated mounting base that work together for superior impact resistance and rebound.

All city posts shall be capable of sustaining a minimum of direct wheel-over impacts at 70 MPH (112 km/h) without damage to the posts or as approved equivalent by the City.

The City Post SM Surface Mount shall consist of system comprised of a tubular marker, a top cap and an integrated mounting base assembly. The tubular marker portion shall consist of round post a minimum of 3.15” in diameter, with 0.125” thick walls (minimum). All posts shall be constructed of UV-stabilized thermo-plastic polyurethane or approved equivalent by the City. The tubular marker mounting base assembly portion shall be permanently bonded to the mounting base assembly. The post assembly shall be affixed to the roadway with Dewalt Screw Anchor, Mfr. Model # PFM1411300 or approved equivalent

The tubular marker top cap portion shall also be permanently bonded to a polyurethane top cap. The Top Cap shall be provided with name of the manufacturer and a plurality of holes to allow the escapement of air when impacted and prevent the collection of rubbish or debris into the tubular marker portion of the post.

Subsection 612.03 shall be revised to include the following:

Spacing, location, color of reflectors, color of tape and placement of tubular markers shall be as shown on the plans.

Subsection 612.04 shall be revised to include the following:

Tubular Markers will be measured by the actual number of the various types installed and accepted.

Subsection 612.05 shall be revised to include the following:

Payment will be made under:

Pay Item	**Pay Unit**
Tubular Marker (SM) (Surface Mount)	Each

Add Hardended Centerline spec, include anchor specs from VIc

include other RRFB specs from CTN-2 (attached) for ped push button post assembly, and ped push button and instruction sign

REVISION OF SECTION 614
TRAFFIC CONTROL DEVICES

of the standard specifications is hereby revised to include the following:

Subsection 614.02 shall include the following:

Construction materials shall be as required per City and County of Denver Traffic Signal, Sign, and Pavement Marking Standards for sign posts, drawings 16.2.5 to 16.2.12.

Subsection 614.08(g) shall include the following:

Pedestrian Push Button shall be installed on a Pedestrian Push Button Assembly Post, as indicated on the plans. Pedestrian Push Button (Wireless) and Instruction sign includes the following:

- the Pedestrian Push Button indicated on the City and County of Denver's Standard Detail 7, Standard drawing No. 16.1.14 (for installation on a Pedestrian Push Button Assembly Post).
- R10-25 sign in lieu of R10-3E sign depicted in Standard drawing No. 16.1.14 (for installation on a Pedestrian Push Button Assembly Post).
- TAPCO LED Bulldog Pedestrian Push Button
- TAPCO Blinker Beam Wireless Radio

All other elements on drawings 16.1.25.1 and 16.1.14 shall be paid for separately.

Rectangular Rapid Flashing Beacon shall conform to the requirements of the City and County of Denver's Standard drawing No. 16.1.25.1. Pedestrian Push Buttons and Instruction Sign shall be paid for separately.

The following manufacturer systems are acceptable: RTC Manufacturing Solar Powered Pedestrian Crossing System with Rectangular Rapid Flashing Beacon, Carmanah Cabinet-Based Rectangular Rapid Flashing Beacon SC315-G, and TAPCO Pedestrian Crosswalk System: Rectangular Rapid-Flashing Beacon.

Subsection 614.10 shall include the following:

Construction shall be as required per City and County of Denver Traffic Signal, Sign, and Pavement Marking Standards for sign posts, drawings 16.2.5 to 16.2.12.

Subsection 614.13 shall include the following:

Pedestrian Push Button shall be measured by the number of units installed and shall include push button, sign, wireless radio, brackets for mounting to the pole, and all work necessary to complete the item.

Steel Sign Post (2x2 Inch Tubing) shall be paid by linear foot and include all work and materials required for installation, including all required in-ground and surface mounting components/foundations.

Rectangular Rapid Flashing Beacon shall be measured by the number of units installed and shall include foundation, Pelco pole, pedestrian crossing sign, rapid flashing bar with amber LED lights, supplemental

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REVISION OF SECTION 614
TRAFFIC CONTROL DEVICES

yellow arrow plaque, pole-mounted radio network controller, solar panel energy source, foundation, and all work necessary to complete the item.

Pelco poles may require the contractor to hydrovac the holes. The resulting slurry of water, soil, and organic material shall be disposed of per XO123. The disposal of material shall not be paid for separately but shall be included in the cost of work.

Traffic Signal Face (12-12-12) shall be measured and paid for by the number of units installed, and shall include all labor and materials required to make operation including brackets and wiring. LED signal lenses shall be included in the cost of the item for Traffic Signal Face and will not be paid for separately.

Subsection 614.14 shall include the following:

Payment will be made under:

Pay Item	**Pay Unit**
Sign Panel (Class I)	Square Foot
Steel Sign Post (2x2 Inch Tubing)	Linear Foot
Signal Head Back Plates	Each
Traffic Signal Face (12-12-12)	Each
Pedestrian Push Button	Each
Rectangular Rapid Flashing Beacon	Each

REVISION OF SECTION 626
MOBILIZATION

Section 626 of the Standard Specifications is hereby revised for this project as follows:

Subsection 626.01 Description is revised to include the following:

Throughout the duration of the project, the Contractor shall limit all of their actives and disturbed areas to Right-of-Way, temporary easements, and permanent easements. All disturbed or damaged areas outside of these limits shall be restored in accordance with the general contract conditions to original or better condition to the satisfaction of the Project Manager at no additional cost to the project.

The contractor and City will work cooperatively to ensure all required forms and material tests are completed satisfactorily to CDOT's requirements. The City reserves the right to delay progress and/or final payment until all daily, weekly, monthly, and annual forms are satisfactorily completed by the contractor and/or adequate information is provided to the City to complete.

If multiple mobilizations are anticipated due to the phasing of the project, the Contractor shall include the requirements of the construction phasing in the bid price for Mobilization. No separate payment will be made for multiple mobilizations due to the Contractor's scheduling. Should the Contractor fail the satisfy any of the above requirement, payment to the Contractor will be withheld or deducted.

This item also includes demobilization of equipment and supplies from this project site.

Subsection 626.02 Basis of Payment shall be revised to include the following:

Payment will be made under:

Pay Item	Pay Unit
Mobilization	Lump Sum

This item shall include the preparation of any plans or documents prior to the preconstruction meeting (including material process control, stormwater management, and traffic control). Additionally, any temporary fencing determined to be necessary by to protect temporary construction easements shall not be measured and paid for separately but shall be included in the work of mobilization.

REVISION OF SECTION 626
PUBLIC INFORMATION SERVICES

make sure to add coordination with Steele Elementary due to proximity to corridor

Subsection 626 of the standard specifications for this project to include the following:

DESCRIPTION

Subsection 626.01 shall include the following:

(a) The Contractor shall provide the following public information services on an ongoing basis throughout the duration of the project:

At the preconstruction conference the Contractor shall introduce the Public Information Manager (PIM) for the project and present a Public Information Plan (PIP) that presents a strategy and methods to lead communication project activities.

The Contractor's PIM shall be a Public Relations or Communications professional, with experience in outreaching multimodal transportation construction projects. The Engineer, after consulting with the City's Public Relations Manager, will approve the Contractor's PIM prior to the preconstruction conference. The identity of the PIM and the PIM'S qualifications shall be submitted to the Engineer fourteen calendar days in advance of the preconstruction conference.

(b) The PIP shall include, but not be limited to the following sections:

a. Outreach Overview – Identify DOTI Public Outreach Leads (POLs), how outreach will be coordinated with the contractor, and plan to interface with POLs and Engineer.

b. Project Locations – List of outreach services will be provided and where, along with map.

c. Outreach Services

d. Website

e. Photos/Videos

f. Media Coordination

g. Lane Closure Reports

h. Construction Signs

i. Translation Services.

A draft of the Public Information Plan will be provided to the Engineer and Public Outreach Leads for review and comment, and then will be finalized by the PIM. The PIM shall update the PIP as needed throughout the duration of the construction period.

(c) Activities of the PIM. The PIM duties are:

a. Outreach Overview

i. The PIM shall be available on every working day, accessible and on call by cell phone at all times, and available upon the request of the Engineer/POLs during times outside of regular working hours. The PIM shall communicate with the Engineer/POLs daily.

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REVISION OF SECTION 626
PUBLIC INFORMATION SERVICES

b. Outreach Services

i. Progress Meetings – At a minimum, weekly progress meetings shall be held with more frequent meetings, as needed, to address specific topics. Meeting minutes shall be provided for each meeting, listing actions and respective responsible parties.

~~ii. Email account – An email account with a DOTI-approved email address shall be created and maintained by PIM. PIM shall be responsible for maintaining email lists using Constant Contact and handling all email communications. Emails shall be responded to within 3 calendar days.~~

c. Stakeholder Coordination – A stakeholder list for each project will be provided by the DOTI Plan Team based on outreach during plan and design phases and updated, as necessary, by the PIM. This stakeholder list will include city council, property owners, businesses and individual residences, commuters, local government entities, neighborhood associations, emergency response agencies, utilities, RTD, schools/universities, and all other stakeholders that are directly adjacent to and affected by the project. The PIM shall be responsible for owning and maintaining the stakeholder list. Stakeholders shall be updated preceding and during construction and contact with stakeholders may be required daily, weekly, monthly or otherwise as needed. Specific stakeholder engagement activities shall include, but are not limited, to:

One round

i. E-Newsletters – developed to update community about projects up to 2x per month.

do we need the two rounds of mailers? Usually only need one.

ii. Mailers/Door Hangers – ~~Two rounds~~ of outreach will be included ~~for each project.~~ Mailers or door hangers shall be delivered to all properties along and within a half-block distance in each direction of each intersecting street. The PIM shall deliver the ~~first round~~ two weeks prior to construction commencement. ~~The PIM shall deliver the second round 48 hours prior to work initiating on each block.~~ DOTI will provide a template for each communication, ~~which will be customized to each corridor indicating bikeway type~~, specific project impacts, and include a map of the project corridor, as well as other pertinent information.

nclude construction yard signs o be set up at major ntersections along the corridor.

iii. Contractor or PIM shall be responsible for all printing and delivery services.

update

d. Website – The PIM shall ~~develop~~ website content for each project indicating project milestones and provide the content to the POLs to be posted to DOTI website. The project website content shall be updated biweekly by the PIM and sent to the DOTI POLs to post to the DOTI website.

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REVISION OF SECTION 626
PUBLIC INFORMATION SERVICES

e. Photos/Videos – The PIM shall take pictures and videos ~~of each corridor~~ to capture project progression. The PIM shall organize media ~~by corridor~~ and provide to POL. The PIM shall incorporate media into the website content and project updates.

f. Media Coordination – The PIM shall ~~draft and finalize press releases, furnish other~~ materials necessary to address media inquiries, ~~and alert the media regarding traffic advisories and media releases for construction efforts.~~ The PIM shall not distribute information to the media without approval by the POL.

g. Lane Closure Reports – Within 24 hours of the weekly coordination meetings, the Contractor superintendent or PIM shall submit weekly lane closure reports using the template provided by the Engineer for posting on the project website. The communication shall include the description of work, lane restrictions, a detour map if warranted, the anticipated start and completion dates, and hours of operation and work schedule.

h. Construction Signs – The Contractor shall erect construction traffic signs with the dates the Contractor expects to initiate and complete construction and with the Contractor's public information office's or PIM's phone number at each major approach to the project. The construction signs shall conform to the requirements of Section 630 and shall be erected at least one week prior to the beginning of construction. Two (2) additional construction signs shall be placed on each corridor that include the name of the project, project website, project email address, and project hotline number. Construction sign size, color, material, reflectivity, and similar requirements shall adhere to Section 630.

i. ~~Translation Services – The PIM shall provide a translator to speak with residents who need to communicate in languages beyond English. The PIM shall translate all newsletters and printed collateral into Spanish, and into other languages as needed.~~

j. Portable Message Sign Panels – The portable message sign panel shall be on the project site 14 calendar days prior to the start of active travel pattern change (i.e. ~~traffic diverters~~) and shall remain in place for 7 calendar days at the beginning of construction and an additional 2 weeks after construction traffic control is removed and the improvements are open to active traffic. Portable message signs shall be placed for each direction of traffic in advance of the traffic pattern change. Portable message sign panel location and duration shall be at the discretion of the Engineer.

i.e. lane shifts, lane closures, etc.

All communications shall include contact information, PIM's name, office phone, City and County of Denver Web-site address with City and County of Denver logo. Cell phone numbers and e-mail addresses shall be provided where service is available.

The Public Information Services Contact Sheet shall include the following:

Owners:

update to Dallas Howell

City and County of Denver
Name: Gavin Weis, Engineer
Address: 201 West Colfax Avenue, Department 506, Denver, CO 80202
Phone: 720-538-6213 Email: Gavin.Weis@denvergov.org

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REVISION OF SECTION 626
PUBLIC INFORMATION SERVICES
BASIS OF PAYMENT

Other Contacts:

Denver Fire Department
Phone: 720-913-3473 (Headquarters) 720-913-3421 (Division 1 – Operations)
Denver Police Department
Phone: 720-913-2000
Denver Public Works Department:
Phone: 720-913-1311
Colorado State Patrol
Phone: 303-239-4501 (Denver Regional Communication Center)

Subsection 626.02 shall include the following:

This is a simple project. Do not need amoritized shcedule of payments.

The Engineer will monitor the PIM and all public information services. When the Contractor provides acceptable public information services in accordance with these specifications, partial payments for the pay item Public Information Services will be made as the work progresses. These partial payments will be made as follows:

When 5 percent of the original Contract amount is earned, 25 percent of the amount bid for this item will be paid.

When 10 percent of the original Contract amount is earned, 40 percent of the amount bid for this item, less all previous payments, will be paid.

When 25 percent of the original Contract amount is earned, 50 percent of the amount bid for this item, less all previous payments, will be paid.

When 75 percent of the original Contract amount is earned, 75 percent of the amount bid for this item, less all previous payments, will be paid.

When 100 percent of the original Contract amount is earned, 100 percent of the amount bid for this item, less all previous payments, will be paid.

Failure to provide acceptable public information services will result in withholding of progress payment for this item. Continued failure to provide the services required will result in non-payment of the corresponding percentage of the original bid item and may result in suspension of the work in those areas affected until acceptable public information services are provided by the Contractor.

For the purpose of public information services, the term “original Contract amount” as used above, shall mean the amount bid for the construction items on this Contract, not including the amounts bid for Public Information Services and Mobilization.

Payment for Public Information Services will be full compensation for all fliers, public information office, telephone lines, and all other labor and materials required to complete the item. Construction Signs will not be measured and paid for separately but shall be included within the work. Portable Message Sign

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REVISION OF SECTION 626
PUBLIC INFORMATION SERVICES
BASIS OF PAYMENT

Panels signs will be paid in accordance with Section 630.

Payment will be made under:

Pay Item	**Pay Unit**
Public Information Management	Lump Sum

REVISION OF SECTION 627
PAVEMENT MARKING

Section 627 of the Standard Specifications is hereby revised for this project as follows:

Subsection 627.02 shall include the following:

Preformed Thermoplastic Pavement Marking (Decorative Green) shall be installed per manufacture's recommendation and as shown in the plans. Contractor shall confirm color with the engineer prior to procurement.

Three-Foot-Long Rubberized Curb for Traffic Circle shall be installed per manufacture's recommendation and as shown in the plans.

Modular Rubber Speed Bump and Modular Rubber Speed Bump End Cap shall be premium recycled rubber speed bumps, Product SKU # TSB-01 as listed on TreeTopProducts.com, or an approved equal. This product shall be colored yellow, NCHRP-350 accepted, MASH 2009 accepted, and resistant to ultraviolet light, ozone, and hydrocarbons.

Subsection 627.11 is hereby deleted and replaced with the following:

627.11 Raised Pavement Markers. Raised pavement markers, which include rubberized parking blocks, rubberized curb, and speed bumps shall be installed at locations where specified in the Contract and in accordance with the manufacturer's recommendations.

Subsection 627.12 shall include the following: Raised pavement markers (Rubberized Parking Blocks, Modular Rubber Speed Bump, and Modular Rubber Speed Bump End Cap) will be measured as a unit in place and shall include all adhesive necessary for installation. All required fasteners, epoxy, and any other implements required for installation shall be included in the work.

Hardened Centerline. Hardened Centerline shall consist of furnishing and installing elements per these specifications and the respective manufacturers recommendations. The Hardened Centerline shall include the following elements: Bump Section End Cap, Bump Section, Hardened Centerline Base (Quantity of 4), Hardened Centerline End Cap (Quantity of 2). Hardened Centerline shall include all materials and equipment necessary for anchoring and installation. Flexible posts shall be paid for separately from Hardened Centerline, as shown on the plan set.

Subsection 627.13 shall include the following:

Pay Item	Pay Unit
Modified Epoxy Pavement Marking	Gallon
Preformed Thermoplastic Pavement Marking (Word/Symbol)	Square Foot
Preformed Thermoplastic Pavement Marking (X-walk Stop line-Yield Line)	Square Foot
Preformed Thermoplastic Pavement Marking (Decorative Green)	Square Foot
Hardened Centerline	Each
3' Long Rubberized Parking Blocks for Raised Median	Each
Modular Rubber Speed Bump	Each
Modular Rubber Speed Bump End Cap	Each

Contractor shall repair any damage to raised pavement marking elements between the time of installation and the final completion of the work regardless of the cause at their own expense.

REVISION OF SECTION 630
CONSTRUCTION ZONE TRAFFIC CONTROL

Section 630 of the CDOT Standard Specifications is hereby revised for this project as follows:

Subsection 630.01 shall include the following:

The key elements of the Contractor's Traffic Control Plan (TCP) and Method of Handling Traffic (MHT) are outlined in Subsection 630.09.

Components of the TCP and MHT for this project are included in the following documents in descending order of precedence:

1. These Special Provisions (Section 630 of the Standard Specifications, and Traffic Control Plan General);
2. CDOT Standard Plan S-630-1, Traffic Controls for Highway Construction;
3. Manual on Uniform Traffic Control Devices (MUTCD) (2009 Edition); and
4. Guide for Work Area Traffic Control, American Traffic Safety Services Association (ATSSA).

Contractor shall submit, in writing, the proposed Method of Handling Traffic (MHT) for review and approval of the Engineer. MHT plans shall be developed in accordance with this section and the construction plans.

Construction materials shall be as required by the Engineer and include two (2) orange flags and mounting devices for the sign panel as designated in the plans.

Subsection 630.05 shall include the following:

Steel drum channelizing devices shall not be used for traffic control.

Subsection 630.06 Channelizing Devise (fixed) shall include the following:

Non-metallic drums or tubular markers may be substituted for vertical panel channelizing devices.

Subsection 630.07 shall include the following:

Non-metallic drums or tubular markers may be substituted for vertical panel channelizing devices.

Subsection 630.09 shall include the following:

STOP/SLOW sign paddles for flaggers shall be 18 inches with letters six inches high.

The Contractor shall not have construction equipment or materials in the lanes open to traffic any time, unless approved by the Engineer.

The Contractor and subcontractors shall equip their construction vehicles with flashing amber lights. Flashing amber lights on vehicles and equipment shall be visible from all directions.

The Contractor shall install construction traffic control devices in locations where they do not block or impede other existing traffic control devices, or sidewalks for pedestrians, disabled persons, or bicyclists. All on-site contractor personnel including superintendents are required to wear appropriate colored, reflective safety vests.

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REVISION OF SECTION 630
CONSTRUCTION ZONE TRAFFIC CONTROL

Subsection 630.10 is hereby revised to include the following:

MHT and TCP

Submission of MHT and/or TCP to the Engineer is required for this project as per the Standard Specifications. Discrete and separate MHT and/or TCP submittals are required for each of the proposed phases of work. When a different MHT is required for a subsequent construction phase, it must be submitted at least two weeks prior to starting that phase.

All proposed MHT shall be approved, in writing, by the Engineer following approval of the Denver Transportation and Mobility Engineering (TME) Department.

Approval of the proposed MHT does not relieve Contractor of liability specifically assigned to it under the contract. Contractor shall erect and maintain warning lights, signs, barricades, and sufficient safeguards around all excavations, embankments, and obstructions. Failure of Contractor to submit detailed, comprehensive, and clearly legible TCP and/or MHT will constitute grounds for rejection of TCP and/or MHT without extensive elaboration or comment from the City.

MHT submittals shall include a tabulation of the anticipated number of hours required for uniformed traffic control. The Contractor shall submit required MHTs and all appropriate Traffic Control Supervisor and Flagger documentation to the Engineer for approval prior to setting any traffic control device.

If the Contractor requires a temporary full closure of any city street, all such requests are subject to the approval of the Engineer. Each full closure request shall be made at least two weeks in advance of the time the full closure is to be implemented.

All proposed lane closures are subject to the approval of the Engineer, and the TME Department. Requests for such lane closures shall be submitted with appropriate MHT at least 5 business days in advance of the time the lane closure is to be implemented. Lane closures will not be allowed to remain unless utilized in continuum for the duration of each working period. Contractor shall make all reasonable efforts to fashion its lane closure proposal to close a minimum number of lanes at any given time.

Certifications

The contractor's superintendent and all others serving in similar supervisory capacity shall have completed two-day Traffic Control Supervisor training as offered by the CCA. The one-day ATSSA Traffic Control Technician (TCT) training along with the two-day ATSSA Traffic Control Supervisor training will serve as an alternate. If the alternate is chosen, the Contractor shall provide written evidence that 80% score was achieved in both training classes. The certifications of completion or certifications of achievement for all appropriate staff shall be submitted to the Engineer at the preconstruction conference.

All flaggers used on this project shall be certified. Copies of flagger certifications shall be submitted to the Engineer.

Permitting and Coordination

Contractor shall obtain all required access and construction permits from the City prior to initiating work along City rights of way. Contractor's project superintendent shall have copies of applicable permits in-hand at all times while on site.

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REVISION OF SECTION 630
CONSTRUCTION ZONE TRAFFIC CONTROL

Contractor shall maintain frequent communication with the Engineer regarding all aspects of the daily and weekly work schedule.

Contractor shall coordinate and cooperate fully with the City & County of Denver, governmental agencies, utility owners and other contractors; to assure adequate and proper traffic control is provided at all times. The Contractor shall coordinate and cooperate fully with any others providing traffic control for other operations in the vicinity of the work to ensure that work or traffic control devices do not interfere with the free flow of traffic except as allowed by the approved MHT.

Subsection 630.13 shall include the following:

General Requirements

During construction of this project, traffic shall use the present traveled roadway at all times unless otherwise directed or authorized by the Engineer.

Contractor shall provide predictable and regular movements for vehicular, bicycle, and pedestrian traffic at all times, unless otherwise authorized or directed by the Engineer.

Contractor shall not have construction equipment or materials in the lanes open to traffic at any time unless otherwise approved by the Engineer. All personal vehicle and construction equipment parking is prohibited where it conflicts with safety, access or the flow of traffic. Personal vehicles and construction equipment parking is prohibited in all private parking lots without payment or the respective property owner's permission.

Vertical cuts or fills greater than 1 inch resulting from construction operations adjacent to traffic lanes or within the clear zone shall be temporarily sloped at a 6:1 or flatter grade and delineated at 35-foot (or more dense) intervals immediately after cutting or filling operations to help safeguard the traveling public.

Construction equipment used on this project shall meet the same minimum exhaust requirements as those specified by the manufacturer of the equipment.

Excavations or holes shall be backfilled or fenced when unattended. During non-construction periods (weekends, holidays, etc.), all work shall be adequately protected to provide for the safety of vehicular, bicycle and pedestrian traffic, as detailed in the contractor's approved TCP and/or MHT.

Location, Positioning and Maintenance of Equipment and Devices

Contractor shall not have construction equipment or materials in the lanes open to traffic any time, unless approved by the Engineer.

Contractor shall install construction traffic control devices in locations where they do not block or impede other existing traffic control devices or sidewalks for pedestrians, disabled persons or bicyclists.

Construction signing shall be removed unless work is in progress, work requires a lane shift, or construction equipment has been left on the job site.

All portable signs shall be removed from the roadway at the end of each work shift.

Portable signs in storage shall not be visible to traffic. All storage areas shall be approved by the Engineer.

All advanced construction signing shall be installed prior to any construction activity and remain in place for all work periods. At intersections where the traffic control is designated to change (for example, from two-way stop to all-way stop, or to all-way yield at a traffic circle), intersection control regulatory signs

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REVISION OF SECTION 630
CONSTRUCTION ZONE TRAFFIC CONTROL

shall be installed prior to, or within 48 hours of installation of pavement markings. At locations along designated bike lanes (striped bike lanes and/or protected bike lanes) where on street parking restrictions are designated to change, parking restriction signs shall be installed prior to, or within 48 hours of installation of pavement markings.

Resetting construction signs during construction as required by the work is considered incidental to the project and will not be paid separately.

Masking of existing or temporary signs may be required by the work is considered incidental to the project and will not be paid for separately.

All personal / employee vehicle and construction equipment parking is prohibited when it conflicts with safety, access, or the flow of traffic.

Traffic control devices used on this project shall be kept clean and in good working order at all times. The Engineer shall make the ultimate determination regarding the explicit definition of "good working order."

Existing signs, traffic signals, or other City-owned traffic control devices damaged due to Contractor's operations shall be replaced in-kind or repaired by the Contractor at no additional cost to the project.

Road Work Ahead advance signs shall be reset as required to match the location of the work zone.

All construction traffic control devices including signs installed under this project shall be installed as depicted in the MUTCD and CDOT M & S Standards.

At locations where such traffic is normally allowed, Contractor shall maintain continuous access through the project for pedestrians, bicyclists and disabled persons. When the existing access route is disrupted by construction or construction-related activities, a temporary access shall be provided. All pedestrian access shall be delineated through the work zone using proper channelizing devices.

During non-construction periods (weekends, holidays, etc.) all work shall be adequately protected to ensure the safety of vehicular, pedestrian and bicycle traffic, as detailed in the Contractor's MHT.

Excavations, drop-offs or holes shall be backfilled, fenced, or barrier-separated when unattended. Costs of removing, resetting, or reinstalling construction traffic control devices shall be considered included in the work and will not be measured and paid for separately.

Access

Contractor shall maintain access to all roadways, side streets, walkways, alleyways, driveways, hike/bike paths and major building entrances at all times unless otherwise approved by the Engineer.

Parking areas temporarily disturbed by construction activities shall be restored to a usable condition during non-working hours. Such temporary parking shall utilize an all-weather surface. The Contractor shall develop an Access Maintenance Plan (AMP) in coordination with, and based on the requirements of, the affected property owners and tenants, and submit same to the Engineer for approval prior to commencement of work. The plan shall detail all barricades, ramps, signs, and temporary means of access required by the property owners or tenants. Prior to commencing any work that affects property access, the AMP for that property shall be submitted to, and approved by, the Engineer.

The AMP shall be coordinated with all affected owners and tenants. The AMP shall include documentation of this coordination, including the approval signatures of the affected owner or tenant. The owner's signature (or owner's representative or Home Owners Association representative) signature shall be considered sufficient approval for temporary access changes for multiple unit apartment or

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REVISION OF SECTION 630
CONSTRUCTION ZONE TRAFFIC CONTROL

condominium buildings. Should the Contractor be unable to obtain approval and signatures, documentation of the effort made to obtain said approval and signatures must be submitted. All access shall be maintained on surfaces equal to or exceeding those in place at the time the access is first disturbed. For short periods of time only as allowed by the Engineer, access may be maintained on an aggregate base course surface.

Contractor shall maintain continuous access throughout the project for pedestrians, bicyclists and disabled persons. When the existing access route is disturbed by construction, a temporary all-weather access shall

be provided. All temporary access shall be a minimum of 5' wide and meet Americans with Disabilities Act (ADA) requirements. Acceptable all-weather surfacing shall be concrete or asphalt construction or as approved by the Engineer. Delineation of pedestrian access through the work zone shall be accomplished using temporary plastic fencing.

Access to individual properties shall be maintained at all times unless otherwise approved by the Engineer.

Access to driveways shall be maintained at all times during construction. The Contractor shall coordinate driveway work with the property owner.

Appropriate signage shall be provided alerting all motorists leaving driveways that enter a work zone as to which direction the through lane is traveling and what access restrictions exist, if any.

Costs incurred by the Contractor to maintain vehicular and pedestrian access will not be paid for separately.

Noise Control

Exempted hours for construction in the City & County of Denver are 7:00am to 9:00pm Monday through Friday and 8:00am to 5:00pm on Saturdays and Sundays per Sections 36-6(B)(7) and 36- 7(5)A, B and C of Denver's Noise Ordinance, Chapter 36 "Noise Control", Denver Revised Municipal Code (DRMC).

If there is an anticipated need to work outside of the exempted hours of construction:

1. The Contractor will need to make a request for a nighttime noise variance as allowed for in Section 36-7(5)C of the DRMC, and
2. The variance process needs to be started a minimum of two to three months prior to the desired start of any work needing to occur outside the exempted hours.

Any noise variance questions should be directed to Paul Riedesel, City Department of Environmental Health, Denver Community Noise Program (T: 720-865-5410; Field Cell:303-916-7967; Fax: 720-865-5532) a minimum of three months prior to the start of the project.

Restricted and Allowable Work – General

The Contractor shall perform all the work on the roadway between the hours of 8:30 A.M. and 3:30 P.M. or as approved by the Engineer. Weekend and nighttime work will be allowed with the prior written approval of the Engineer.

Work will not be permitted that directly or indirectly interferes with the flow of traffic between the hours of 5:30 AM and 8:30 AM Monday through Friday; between the hours of 3:00 PM to 6:30 PM Monday through Thursday; and after 2:00 PM on Fridays unless otherwise authorized by the Engineer.

Night-time work will be allowed on this project. All night work requires the prior approval of the Engineer as well as an approved variance to the Noise Ordinance.

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REVISION OF SECTION 630
CONSTRUCTION ZONE TRAFFIC CONTROL

Weekend work shall be allowed as directed or authorized by the Engineer. Requests for weekend work shall be submitted to the Engineer by no later than close of business on Wednesday preceding the weekend during which the Contractor desires to work.

The contractor is allowed to work on multiple work areas, streets, blocks or intersections simultaneously if that work schedule is so approved by the Engineer.

Wherever other laws, ordinances, regulations or orders are more restrictive, they shall take precedence over these requirements.

Holidays

Work that interferes with traffic on any day of a holiday; or 3-day or 4-day holiday weekend shall generally not be permitted.

Holidays shall be as defined in Commencement and Completion of Work. Work after 12:00 noon on the day preceding such a holiday weekend is similarly included in the restricted times. Exceptions to these requirements must be approved by the Engineer and Denver TME.

Special Events and Other Considerations

The City may place restrictions on allowable work periods during specific heavily-traveled periods. The Engineer will notify the Contractor of any such restrictions during the course of normal coordination for this project.

Any proposed work on the roadway during all City-defined special events shall be coordinated with the Engineer.

Subsection 630.17 is hereby deleted and replaced with the following:

Contractor shall furnish all personnel and materials necessary to perform the Construction Zone Traffic Control as required by the approved MHT. Except as noted within this Project Special Provision, these quantities will not be measured.

Placement and removal of temporary pavement markings for traffic control during applicable construction phases will not be measured and paid for separately but shall be included in the work.

All MHT, TCP and AMP preparation and submittal shall be included in the cost of traffic control items. All rental, purchase, installation, maintenance (including cleaning), resetting and removal of all equipment, signs and other devices is considered included. All costs including electrical power, batteries for all devices, temporary lighting or temporary signals are considered included.

Subsection 630.18 is hereby revised to include the following:

Payment for Traffic Control shall include the Traffic Control Plans, all devices, Traffic Control Inspection, Traffic Control Management, Uniformed Traffic Control, flagging, detours, temporary pavement markings, and all other personnel and resources necessary to complete the work. Payment will be made based on the number of calendar days Traffic Control is in place.

Payment will be made under:

Pay Item	**Pay Unit**
Traffic Control	Lump Sum
Speed Change Warning Flag	Each

TRAFFIC CONTROL PLAN GENERAL

The key elements of the Contractor's method of handling traffic (MHT) are outlined in subsection 630.10(a).

The contractor shall prepare plans for MHT, this work shall not be paid for separately but shall be included in the work of Construction Zone Traffic Control.

The components of the TCP for this project are included in the following:

1. Section 630 of the specifications.
2. Latest revised Standard Plan S-630-1, Traffic Controls for Highway Construction, and Standard Plan S-630-2.
3. Manual on Uniform Traffic Control Devices for Streets and Highways.
4. Transportation Standards and Details for the Engineering Division (City and County of Denver)
5. General Traffic Control Procedures for Street/Public Occupancy Permits (City and County of Denver)
6. Pedestrian Walkway Entrance Requirements (City and County of Denver)
7. Street Cut Entrance Requirements (City and County of Denver)

Special Traffic Control Plan requirements for this project are as follows:

1. This project includes restrictions to work times and days that affect traffic during peak traffic times and days, holidays, holiday eves (all day), holiday weekends, special events, and other circumstances as described later in this special provision. Wherever other laws, ordinances, regulations or orders are more restrictive, they shall take precedence over these requirements.
2. All lane closures are subject to the approval of the Engineer. The Contractor shall submit a plan for each lane closure to the Engineer for review and approval at least 5 business days in advance of the time the lane closure is to be implemented. Lane closures will not be allowed to remain unless being used continuously for the purpose for which they were set up.
3. The Contractor shall coordinate and cooperate fully with the City and County of Denver, governmental agencies, utility owners, and other contractors, to assure adequate and proper traffic control is provided.
4. The Contractor shall coordinate and cooperate fully with any others providing traffic control for other operations to assure that work or traffic control devices do not interfere with the free flow of traffic except as allowed by the Traffic Control Plan. During flagging and uniform traffic control operations at signalized intersections, the signal heads shall be masked.

During the construction of this project, traffic shall use the present traveled roadway unless otherwise approved by the Engineer.

The Contractor shall not have construction equipment or materials in the lanes open to traffic at any time, unless approved by the Engineer.

The Contractor shall provide, erect and maintain proper traffic control devices until the site is open to traffic. Prior to construction, the Contractor shall submit for approval a traffic control plan that complies with Manual of Uniform Traffic Control Devices and City Standards for all activities that affect the roadway and sidewalk.

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TRAFFIC CONTROL PLAN GENERAL

Unless otherwise approved by the Engineer, the Contractor's equipment shall follow normal and legal traffic movements. The Contractor's ingress and egress of the work area shall be accomplished with as little disruption to traffic as possible. Traffic control devices shall be removed by picking up the devices in a reverse sequence to that used for installation. This may require moving backwards through the work zone. When located behind barrier or at other locations shown on approved traffic control plans, equipment may operate in a direction opposite to adjacent traffic.

The Contractor shall be responsible for the provision of a safe travel way on all streets, roadways sidewalks, and trails on and adjacent to the job site. The Contractor shall erect or cause erection of proper traffic control warning devices around all excavations, embankments, and obstructions and shall be responsible for the proper maintenance of said erected devices, in accordance with the street occupancy permit and the MUTCD.

The flow of vehicular, pedestrian, and bicycle traffic on public streets and roadways shall be maintained at all times during construction in accordance with the rules, regulations, and conditions as set forth in the street occupancy permit issued by the City of Denver. Signs, barricades, lights, and warning devices shall be constructed and used in accordance with the MUTCD and the Colorado supplement.

The Contractor shall cause suitable warning lights to be provided and kept lighted at night or other times when visibility is limited. The Contractor shall provide flaggers and/or off-duty police protection as may be determined by the Engineer for the protection of the public, as well as workers on the job site.

The Contractor shall coordinate with the Engineer so that arrangements may be made by the Contractor for detours, parking, and access to property adjacent to work, etc., 48 hours prior to their need. A minimum notification of one (1) week is required when detouring a street.

The Contractor shall not work within any portion of a street without receiving a Street Occupancy Permit from the Traffic Engineer prior to such work. Full roadway closures will be reviewed on a case-by-case basis. The City reserves the right to refuse to allow full road closures. Requirements for such closures will be determined at the time of issuance of permit. The Contractor will be responsible for all public notices, public meetings, and requirements as outlined in the Street Occupancy Permit or by the Engineer.

The Contractor shall be responsible for all damages to the work due to failure to place barricades, signs, lights, flaggers, and other workers to protect it. Whenever evidence of such damage is found prior to acceptance, the Engineer may order the damaged portion immediately removed and replaced by the Contractor.

Sufficient traffic control devices are included in the plans to cover expected construction activities. Should the Contractor elect to utilize additional devices to enhance the operation, the additional devices will not be paid for, but shall be provided at the Contractor's expense unless otherwise approved for payment by the Engineer.

All flagging stations used at night shall be adequately illuminated in accordance with the MUTCD. Adequate illumination of flagging stations shall include the use of light plants whenever other sources of adequate lighting are not available. Adequate illumination shall be as approved or determined by the Engineer. The cost of lighting shall be included in the work.

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TRAFFIC CONTROL PLAN GENERAL

All costs incidental to the foregoing requirements shall be included in the original contract prices for the project.

Lane Closure Restrictions – General

Failure to complete work within the terms of the Street Occupancy Permit may result in a "stop work" order.

All lane closures are subject to the approval of the Engineer. A MHT shall be approved by the Engineer prior to any request for a lane closure. A written request for each lane closure shall be made at least 5 business days in advance of the time the lane closure is to be implemented. Lane closures will not be allowed to remain unless they are continuously utilized for the intended purpose for which they were approved.

City and County of Denver

Weekday work occurring on the roadway shall generally occur between 8:30am and 3:30pm as approved by the Engineer. Project-related work outside the indicated hours is permitted with the prior approval of the Engineer. If the City permit authorizes work outside the hours indicated herein, the language on the permit shall control.

The contractor may be allowed to work weekdays prior to 8:30am and after 3:30pm Monday through Friday if:

1. The work is accomplished during daylight hours;

2. The work takes place off the roadway and does not interfere with the flow of traffic; and

3. The given activity has been approved by the Engineer.

Unless otherwise authorized by the Engineer or explicitly stated in the plans, one lane in each direction must be maintained on all affected roadways.